EXP 2. 8-Queen Problem

Aim

To place eight queens on an 8 × 8 chessboard so that no two queens attack each other.

Algorithm

1. Start with an empty board represented by one queen position per row.
2. Try placing a queen in each column of the current row.
3. Check whether the column and both diagonals are safe.
4. If safe, place the queen and recursively process the next row.
5. If no column works, backtrack and try another position.
6. **When all eight rows are filled, print the board.**

Program

AI.py - C:/Users/Barath/AppData/Local/Programs/Python/Python313/AI.py (3.13.15)

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```
N = 8
board = [-1] * N
def solve(row):
    if row == N:
        return True

    for col in range(N):
        safe = True
        for r in range(row):
            if board[r] == col or abs(board[r] - col) == row - r:
                safe = False
                break

        if safe:
            board[row] = col
            if solve(row + 1):
                return True
            board[row] = -1
    return False

if solve(0):
    for r in range(N):
        print(" ".join("Q" if board[r] == c else "." for c in range(N)))
```

Output

```
>>>
==== RESTART: C:/Users/Barath/AppData/Local/Programs/Python/Python313/AI.py ====
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
>>>
```

Result

Thus, eight queens were placed on the chessboard without any two queens attacking each other.